SPA617

RADIO ASTRONOMY LABORATORY

Experiment 1

Antenna Characterization

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1 Aim

To characterize the given antenna by measuring its beam pattern in co-polarization and cross-polarization configurations.

2 Objectives

1. Determine the operating frequency and return loss from VSWR measurements.
2. Measure the effect of distance on received power.
3. Measure the far-field power pattern versus angle (co- and cross-polarization).
4. Determine the half-power beamwidths and estimate the directivity.

3 Theory

An antenna converts guided electrical power into radiated electromagnetic waves (transmission) and vice versa (reception). A $\lambda/2$ dipole is fed at its centre; a $\lambda/4$ monopole over a ground plane behaves like half of a dipole, the ground plane acting as a mirror. For a monopole, $L = \lambda/4 = c/4f$.

Beam pattern. The radiation pattern consists of a main lobe, side lobes and a back lobe. The *E-plane* contains the electric-field vector and the propagation vector; the *H-plane* contains the magnetic-field vector and the propagation vector; the two are perpendicular.

Reflection coefficient, VSWR, return loss.

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad \mathrm{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{V_{\max}}{V_{\min}}, \quad |\Gamma| = \frac{V_{\max}-V_{\min}}{V_{\max}+V_{\min}}, \quad \mathrm{RL} = -20\log_{10}|\Gamma| \ \mathrm{dB}. \tag{1}$$

Free-space propagation. In the far field the received power follows the Friis relation $P_r = P_t G_t G_r \,(\lambda/4\pi R)^2$, so in dB

$$P_r(\mathrm{dBm}) = \mathrm{const} - 20\log_{10} R, \tag{2}$$

i.e. a straight line of slope -1 in a P_r–$20\log_{10} R$ plot, and a drop of $20\log_{10} 2 \approx 6\,\mathrm{dB}$ when the distance is doubled. The far-field condition is $R \gtrsim 2D^2/\lambda$ (and here $R \geq 3\lambda$ was required).

HPBW and directivity. The half-power beamwidth is the angular width between the $-3\,\mathrm{dB}$ points of the main lobe. With θ_E, θ_H in radians,

$$D = \frac{4\pi}{\Omega_A} \approx \frac{4\pi}{\theta_E \theta_H}. \tag{3}$$

4 Apparatus

Monopole and dipole antennas, RF signal generator, digital storage oscilloscope ($1\,\mathrm{M}\Omega$ input), spectrum analyser, RF Tee connector, $50\,\Omega$ termination, antenna support stand, coaxial cables and adapters. The DUT (receiver) was a directional multi-element antenna (Fig. 4 of the manual); the transmitter was a monopole.

5 Procedure

Part A: signal generator, Tee, DUT and oscilloscope connected as in the manual; around each test frequency a small-span sweep covering one ripple period was made and $V_{\max}$, $V_{\min}$ were noted; VSWR, $|\Gamma|$ and RL were computed.

Part B: the monopole length was set to $\lambda/4$ for the chosen frequency, the antennas were aligned (same polarization and height) and the received power was recorded on the spectrum analyser for several separations.

Parts C–D: at a fixed frequency and separation the receiver was rotated about the vertical and horizontal axes to obtain the H- and E-plane patterns, first with parallel (co-pol) and then with perpendicular (cross-pol) transmitter polarization.

Part E: HPBW from the $-3\,\mathrm{dB}$ points of the co-polar patterns and directivity from the formula above.

6 Observations and Calculations

6.1 Part A: VSWR and return loss

Monopole length used: $L = \dfrac{c}{4f} = \dfrac{3\times10^{8}}{4\times2\times10^{8}} = 37.5\,\mathrm{cm}$ for $f = 200\,\mathrm{MHz}$ ($\lambda = 1.5\,\mathrm{m}$).

Table 1: VSWR and return-loss measurements.

f (MHz)	$V_{\max}$ (mV)	$V_{\min}$ (mV)	VSWR	$\lvert\Gamma\rvert$	RL (dB)
200	412	384	1.073	0.035	29.08
30	430	240	1.792	0.284	10.95

Sample calculation ($f = 200\,\mathrm{MHz}$): VSWR $= 412/384 = 1.073$; $|\Gamma| = (412-384)/(412+384) = 0.0352$; RL $= -20\log_{10}(0.0352) = 29.08\,\mathrm{dB}$.

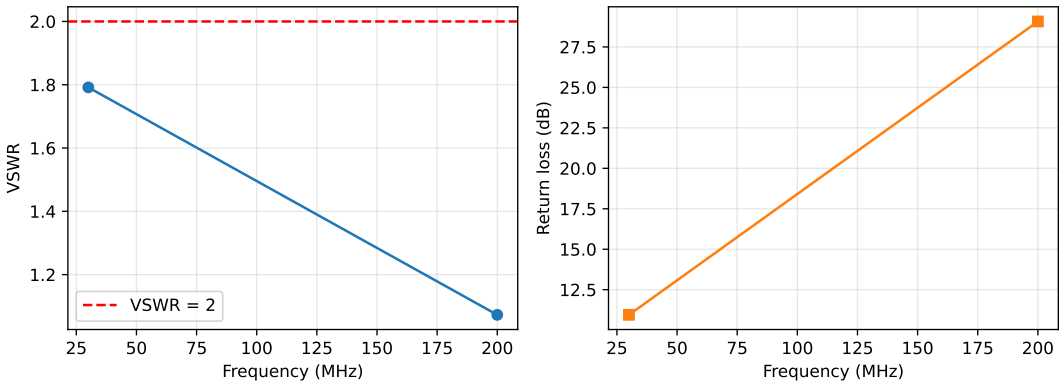


Figure 1: VSWR and return loss versus frequency.

6.2 Part B: Effect of distance

Table 2: Received power versus separation (measurement order was far to near).

R (cm)	P_r (dBm), 10 dB atten.	P_r (dBm), pre-amp on, 20 dB atten.
904	−42.51	−46.80
744	−32.97	−38.77
584	−29.95	−29.27
424	−29.75	−30.90
264	−31.33	−26.96

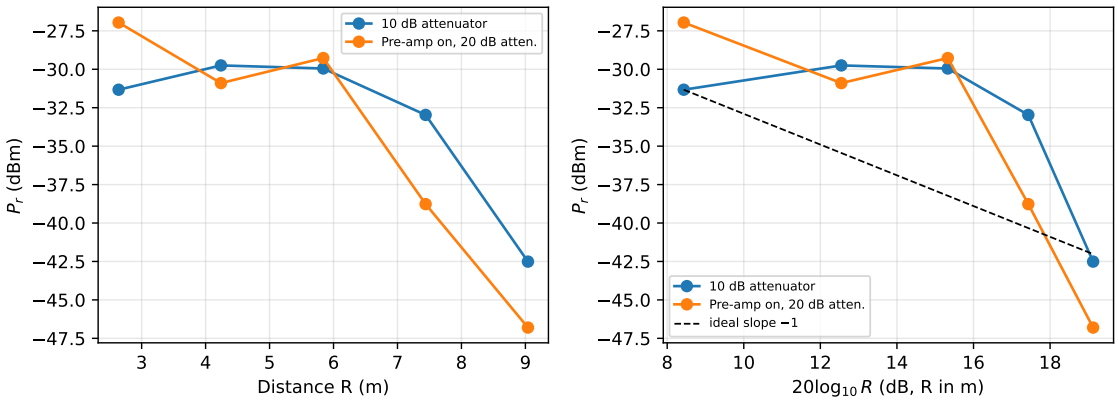


Figure 2: Left: P_r versus R. Right: P_r versus $20\log_{10} R$ with the ideal slope -1 (dashed).

Linear fits of P_r versus $20\log_{10} R$ give slopes of -0.81 ($r = -0.64$) and -1.65 ($r = -0.85$) for the two data sets; using only the three farthest points ($R \geq 4.24\,\text{m}$) the slopes are -1.76 and -2.47.

6.3 Parts C and D: Beam patterns

Angular scans were taken in 10° steps (co-pol) and 20° steps (cross-pol) over 0°–360° (complete tables in Appendix A: Tables 3 and 4).

Table 3: Co-polarization received power, E- and H-plane (10° steps).

Angle (°)	E-plane (dBm)	H-plane (dBm)	Angle (°)	E-plane (dBm)	H-plane (dBm)
0	−27.40	−44.70	190	−52.76	−58.93
10	−27.34	−44.66	200	−41.60	−51.51
20	−27.60	−43.91	210	−41.70	−50.41
30	−28.51	−42.93	220	−40.30	−50.90
40	−29.30	−42.75	230	−40.00	−52.88
50	−30.91	−42.05	240	−40.93	−53.50
60	−32.73	−41.69	250	−42.20	−53.54
70	−34.20	−40.93	260	−39.36	−54.15
80	−36.91	−40.88	270	−38.10	−50.95
90	−41.20	−41.03	280	−36.66	−44.40
100	−41.90	−42.90	290	−34.23	−41.64
110	−40.50	−51.52	300	−32.50	−40.30
120	−41.70	−52.30	310	−31.43	−39.88
130	−39.90	−51.20	320	−30.40	−38.49
140	−39.12	−46.91	330	−30.18	−39.70
150	−41.80	−47.70	340	−28.77	−41.20
160	−46.50	−47.57	350	−28.17	−43.17
170	−46.60	−49.41	360	−27.62	−44.74
180	−46.10	−53.76			

Table 4: Cross-polarization received power, E- and H-plane (20° steps; H-plane value at 300° is uncertain).

Angle (°)	E-plane (dBm)	H-plane (dBm)	Angle (°)	E-plane (dBm)	H-plane (dBm)
0	−62.68	−27.76	200	−58.62	−47.56
20	−50.50	−28.32	220	−56.67	−41.84
40	−48.06	−30.50	240	−56.40	−44.70
60	−48.62	−32.12	260	−52.70	−45.67
80	−48.80	−32.92	280	−43.11	−57.25
100	−52.88	−33.53	300	−43.00	−40.80
120	−49.37	−39.61	320	−44.15	−32.50
140	−43.10	−41.41	340	−47.81	−29.51
160	−54.50	−47.67	360	−67.13	−27.66
180	−58.50	−40.89			

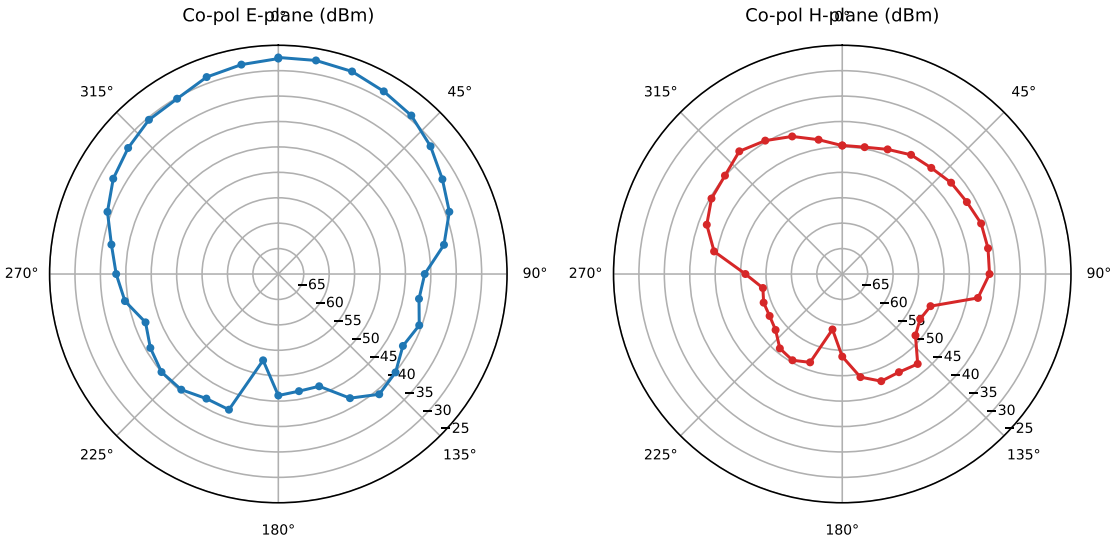


Figure 3: Co-polarization polar patterns (received power in dBm).

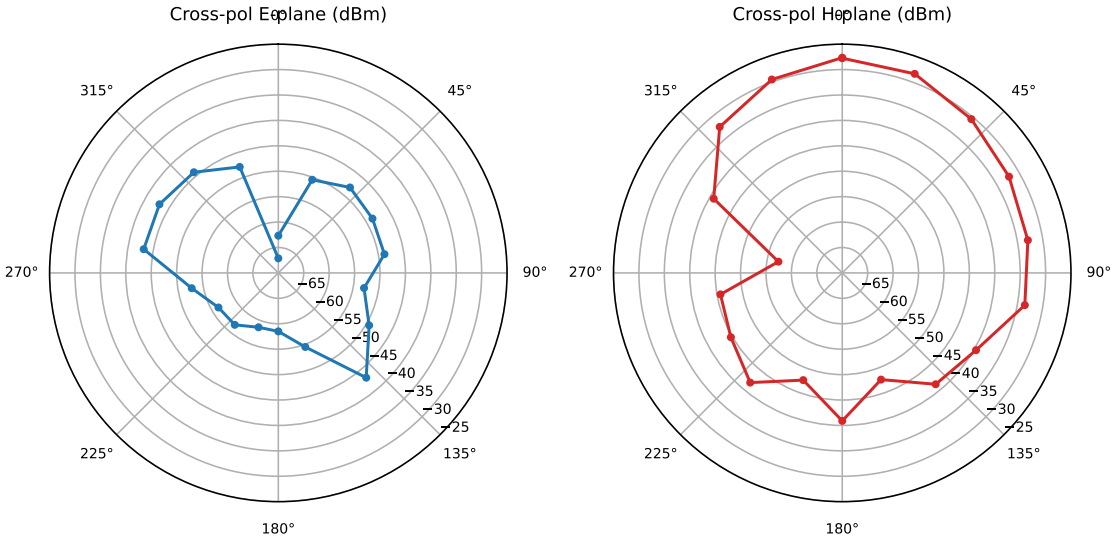


Figure 4: Cross-polarization polar patterns (received power in dBm).

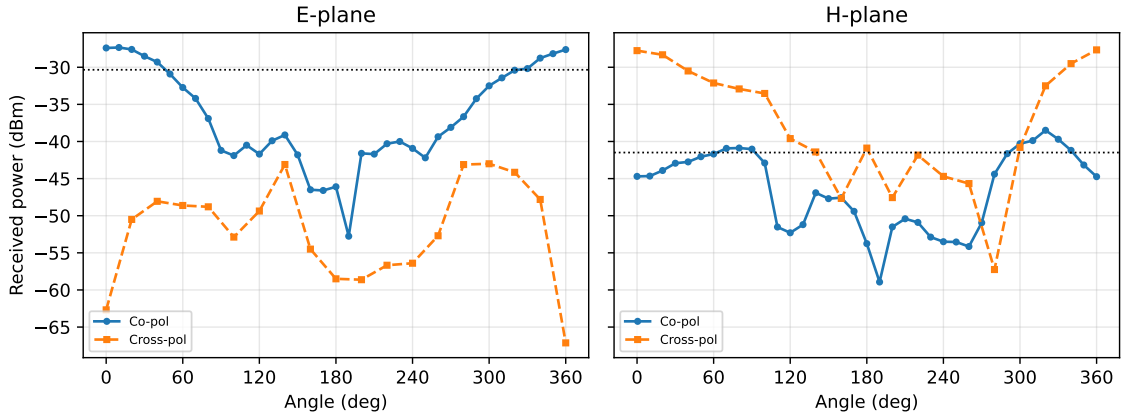


Figure 5: Co- and cross-polar patterns in Cartesian form; dotted line = −3 dB level of the co-polar peak.

6.4 Part E: HPBW and directivity

E-plane: peak = −27.34 dBm (at 10°); −3 dB level = −30.34 dBm. Crossings at ≈ 47.3° and ≈ 36.5° either side of the peak, so $\theta_E \approx 83.7° = 1.461$ rad.
H-plane: peak = −38.49 dBm (at 320°); −3 dB level = −41.49 dBm. Crossings at ≈ 28.9° and ≈ 21.5° either side, so $\theta_H \approx 50.4° = 0.879$ rad.

$$D \approx \frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{1.461 \times 0.879} \approx 9.8 \ (\approx 9.9 \text{ dBi}).$$

7 Results

Quantity	Symbol	Value
Antenna operating frequency	f_r	200 MHz (lowest VSWR of the two points measured)
Minimum VSWR	VSWR_{min}	1.073
Return loss at f_r	RL	29.08 dB
Frequency range for VSWR ≤ 2	–	both measured points (30 and 200 MHz) satisfy it; band edges not resolved (too few points)
E-plane HPBW	θ_E	83.7°
H-plane HPBW	θ_H	50.4°
Estimated directivity	D	9.8 (9.9 dBi)

8 Conclusions

- **Operating frequency, VSWR, RL.** The monopole cut to $\lambda/4 = 37.5$ cm gave VSWR = 1.07 and RL = 29 dB at 200 MHz, i.e. only ≈ 0.1% of the power is reflected ($|\Gamma|^2 = 1.2 \times 10^{-3}$): the antenna is well matched at its design frequency, consistent with the theoretical $\lambda/4$ resonance. The second point gave VSWR = 1.79 ($|\Gamma| = 0.28$, RL = 10.9 dB, ≈ 8% power reflected).
- **Distance dependence.** Ideal free-space behaviour is a −1 slope versus $20 \log_{10} R$ (−6 dB per doubling). The measured data are not monotonic: power rises as the antennas approach from 9 m to 5.8 m and then flattens or falls at 4.2 and 2.6 m. The far points are steeper than −1 (slopes −1.8 to −2.5), while the nearest points ($R < 3\lambda = 4.5$ m) lie in the near-field region where the $1/R^2$ law does not hold.

- **Beam shapes.** The E-plane pattern is a broad main lobe ($\theta_E \approx 84°$) with a back lobe about 19 dB below the peak; the H-plane pattern is narrower ($\theta_H \approx 50°$) with a deep null near 190°. The H-plane maximum occurs near 320° (about 40° off the 0° reference), suggesting a misalignment of the rotation axis/boresight.

- **Co- versus cross-polarization.** In the E-plane the cross-polar level is ≈ 35 dB below the co-polar level at 0° and at least ≈ 16 dB below over the full scan, showing good polarization purity. In the H-plane, however, the cross-polar level at 0° (-27.8 dBm) is *higher* than the co-polar level (-44.7 dBm) by ≈ 17 dB.

- **HPBW and directivity.** $\theta_E = 83.7°$, $\theta_H = 50.4°$ give $D \approx 9.8$ (≈ 9.9 dBi), typical of a small Yagi/log-periodic-type directional antenna. The formula $4\pi/\theta_E\theta_H$ is approximate (it ignores side lobes), so this is an estimate.

- **Variation with frequency.** Patterns were recorded at a single frequency only; the required comparison at two further frequencies was not done. In general the beam narrows (HPBW decreases) as frequency increases because the antenna is electrically larger.

- **Uncertainty sources.** Multipath and reflections from nearby objects; near-field measurement ($R < 3\lambda$); antenna misalignment and pointing/rotation errors; cable and connector movement; spectrum-analyser amplitude accuracy and changing attenuator/pre-amp settings; oscilloscope reading error in $V_{\max}, V_{\min}$ (a 1 mV error changes RL by ~ 0.3 dB at 200 MHz); transmitter power drift; sparse angular sampling (10°–20°).